

CLASS 11 BIOLOGY • CHAPTER 9: BIOMOLECULES

Complete NCERT Detailed Notes (Sections 9.1 to 9.8 Complete Coverage)

Subject: Biology (Code 044) | Unit III: Cell: Structure and Functions

Session 2026–2027 • Board & NEET Syllabus

9.1 How to Analyse Chemical Composition

Biomolecules: All the carbon compounds that we get from living tissues are called biomolecules. Living tissues contain organic compounds along with inorganic elements and salts.

A. Chemical Fractionation (Trichloroacetic Acid Method)

1. Take any living tissue (a vegetable slice or a piece of liver) and grind it in **Trichloroacetic acid (Cl_3CCOOH)** using a mortar and pestle.
2. A thick slurry is formed. Filter this slurry through cheesecloth or cotton cloth.
3. This separates the tissue into two distinct fractions:

1. Acid-Soluble Pool (Filtrate):

- Contains **Biomicromolecules**.
- Molecular weights: **18 to 800 Daltons (Da)**.
- Represents cytoplasmic composition.
- *Examples:* Monosaccharides, Amino acids, Nucleotides, Nucleosides, Inorganic salts.

2. Acid-Insoluble Fraction (Retentate):

- Contains **Biomacromolecules**.
- Molecular weights: **> 10,000 Daltons (Da)**.
- Represents cellular macromolecules and membranes.
- *Examples:* Proteins, Nucleic acids, Polysaccharides, and **Lipids***.

★ NCERT Question — Why are Lipids retained in the Acid-Insoluble Fraction?

Lipids have small molecular weights (**< 800 Da**) and are strictly biomicromolecules. However, lipids are arranged in cell membranes and organelle membranes. When tissue is ground, membranes break and form water-insoluble **vesicles**. These vesicles are not acid-soluble and cannot pass through the cheesecloth, getting collected in the **acid-insoluble retentate**.

B. Ash Analysis (Inorganic Constituents Analysis)

- Weigh a small amount of living tissue = **Wet Weight**.
- Dry the tissue completely in an oven to remove all moisture/water = **Dry Weight**.
- Fully ignite/burn the dried tissue: All carbon compounds are oxidised to gaseous forms (**CO_2 , H_2O** vapour) and escape.
- The remaining residue is called **Ash**, containing inorganic elements (**Ca^{2+} , Mg^{2+} , K^+ , Na^+ , SO_4^{2-} , PO_4^{3-} , Cl^-**).

9.2 Primary and Secondary Metabolites

Primary Metabolites:

- Have direct, identifiable physiological functions in host organisms.
- Play key roles in normal growth, development, and respiration.
- Universally present in all living cells (e.g., Glucose, Alanine, Fatty acids, ATP).

Secondary Metabolites:

- No direct identifiable roles in primary host metabolism.
- Produced mainly by plants, fungi, and microbes.
- Have ecological importance (defence against herbivores, insects, pathogens) or economic value to humans.

NCERT Table of Secondary Metabolites

Category	NCERT Examples	Biological / Human Value
Pigments	Carotenoids, Anthocyanins	Photosynthesis & floral coloration
Alkaloids	Morphine, Codeine	Analgesics (pain relievers), narcotics
Terpenoids	Monoterpenes (Pinene), Diterpenes	Resins, insect-repellent essential oils
Essential Oils	Lemon grass oil, Eucalyptus oil	Fragrance, antibacterial action
Toxins	Abrin (from <i>Abrus</i>), Ricin (from Castor)	Ribosome-inactivating defensive proteins
Lectins	Concanavalin A	Carbohydrate-binding proteins
Drugs	Vinblastin , Curcumin	Vinblastin (anticancer drug), Curcumin (turmeric)
Polymeric Substances	Rubber, Gums, Cellulose	Industrial raw materials, cell wall structural support

9.3 Biomacromolecules

Biomacromolecules: Chemical compounds found in the acid-insoluble fraction of living tissues having molecular weights greater than **10,000 Daltons (Da)**. They are large polymeric molecules formed by polymerization of repeating micromolecular monomer units.

There are four major classes of biomacromolecules found in living systems:

1. **Proteins:** Polymers of amino acids.
2. **Polysaccharides:** Polymers of monosaccharides.
3. **Nucleic Acids:** Polymers of nucleotides.
4. **Lipids:** Though found in the acid-insoluble pool due to vesicle formation, lipids are not polymeric and have molecular weights < 800 Da.

9.4 Proteins & Amino Acids

Proteins: Unbranched **heteropolymers** of amino acids linked together by **peptide bonds ($-CO-NH-$)**. A protein is never a homopolymer because it consists of varying sequences of the 20 standard amino acids.

A. Amino Acid Structure & Chemistry

Amino acids are **substituted methanes** where four substituent groups are attached to the same central **α -carbon**: an amino group ($-NH_2$), a carboxyl group ($-COOH$), a hydrogen atom ($-H$), and a variable side chain ($-R$).

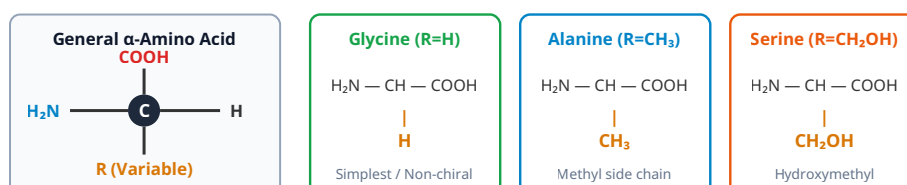


Figure 9.1: General structure of α -amino acids and 3 standard NCERT amino acids.

B. Types of Amino Acids & Zwitterion

- **Neutral:** Valine, Alanine, Glycine, Leucine, Isoleucine.
- **Acidic:** Glutamic acid, Aspartic acid (extra $-COOH$ in R-chain).

- **Basic:** Lysine, Arginine, Histidine (extra $-\text{NH}_2$ in R-chain).
- **Aromatic:** Tyrosine, Phenylalanine, Tryptophan.
- **Zwitterion Form:** At neutral/isoelectric pH, amino acids exist as dipolar ions with net charge zero ($\text{H}_3\text{N}^+ - \text{CH}(\text{R}) - \text{COO}^-$).

★ **Important Biological Proteins (NCERT High-Yield):**

- **Collagen:** Most abundant protein in the **animal world** (intercellular ground substance).
- **RuBisCO:** Most abundant protein in the **entire biosphere**.
- **GLUT-4:** Glucose transporter protein that enables glucose transport into cells.
- **Antibody:** Fights against infectious agents.

9.5 Polysaccharides (Carbohydrates)

Polysaccharides: Long chains of monosaccharides linked by **glycosidic bonds** formed by dehydration condensation between carbon atoms of adjacent sugar rings.

Polysaccharide	Monomer Unit	Structural Architecture	Function & Iodine (I_2) Test
Starch	α -D-Glucose	Composed of Amylose (unbranched helical) + Amylopectin (branched).	Energy storage in plants. Gives Blue color with I_2 (helices trap iodine).
Glycogen	α -D-Glucose	Highly branched polymer. Right end = Reducing ; Left end = Non-reducing .	Energy storage in animals (liver/muscles). Gives Red-Violet color with I_2 .
Cellulose	β -D-Glucose	Straight unbranched linear chains (β-1,4 bonds) packed parallel.	Plant cell walls & cotton fibre. Does NOT trap I_2 (No color) (no helices).
Chitin	N-acetylglucosamine (NAG)	Complex homopolymer of amino-sugar derivatives (β-1,4 bonds).	Exoskeleton of arthropods; fungal cell wall.
Inulin	D-Fructose	Linear homopolymer of fructose (β-2,1 bonds).	Storage in <i>Dahlia</i> roots; used to test renal GFR.

9.6 Nucleic Acids

Nucleic Acids (DNA and RNA): Polynucleotide macromolecules found in the acid-insoluble fraction. Each nucleotide consists of 3 distinct components: a nitrogenous base, a pentose sugar, and phosphoric acid.

1. Nitrogenous Bases:

- **Purines (Double-ring):** Adenine (A), Guanine (G).
- **Pyrimidines (Single-ring):** Cytosine (C), Thymine (T in DNA), Uracil (U in RNA).

2. Pentose Sugar: Ribose (RNA) or 2'-Deoxyribose (DNA).

Terminology Rules:

- **Base + Sugar = Nucleoside:** Adenosine, Guanosine, Cytidine, Thymidine, Uridine.
- **Nucleoside + Phosphate = Nucleotide:** Adenylic acid (AMP), Guanylic acid (GMP), Cytidylic acid, Thymidylic acid, Uridylic acid.
- **Joined by 3'-5' Phosphodiester bonds.**

9.7 Structure of Proteins & B-DNA Architecture

A. Four Levels of Protein Structural Organization

Level	Features & Architectural Organization	Stabilizing Forces & Examples
1. Primary (1°)	Positional sequence of amino acids in a linear polypeptide chain. • Left End: First amino acid = N-terminal (free $-NH_2$) • Right End: Last amino acid = C-terminal (free $-COOH$)	Covalent Peptide Bonds Ex: Insulin sequence.
2. Secondary (2°)	Localized regular folding of polypeptide chain: • α-Helix: Right-handed spiral helix (e.g., Keratin in hair) • β-Pleated Sheet: Polypeptides pleated side by side (Silk fibroin)	Hydrogen bonds between backbone $C=O$ and $N-H$ groups.
3. Tertiary (3°)	Polypeptide chain folds extensively upon itself like a hollow woolen ball into a compact 3D conformation , creating functional pockets (Active Sites). <i>Mandatory for biological activity of enzymes.</i>	Hydrophobic interactions, Disulfide bridges ($-S-S-$), Ionic bonds, H-bonds, Van der Waals. Ex: Myoglobin, Enzymes.
4. Quaternary (4°)	Spatial assembly of two or more independent polypeptide subunits.	Non-covalent hydrophobic & ionic bonds. Ex: Adult Hemoglobin (Hb) has 4 subunits ($2\alpha + 2\beta$ chains).

B. Watson-Crick Double-Helix Model of B-DNA

- Two polynucleotide chains running in **antiparallel directions** ($5' \rightarrow 3'$ and $3' \rightarrow 5'$).
- Sugar-phosphate-sugar backbone on outside; nitrogenous bases project inward.
- **Complementary Base Pairing:** **A = T** (2 H-bonds) and **G $\equiv$ C** (3 H-bonds).
- **Helical Dimensions:** Pitch = 3.4 nm (34 Å); **10 bp per turn**; distance between adjacent bp = **0.34 nm (3.4 Å)**; Diameter = 2.0 nm (20 Å); Angle of rotation per bp = **36°**.

9.8 Enzymes (Biocatalysts)

Enzymes: Biological catalysts composed of proteins with a specific 3D tertiary structure containing specialized catalytic clefts called **Active Sites** into which substrates fit. (Exception: Catalytic RNAs are called **Ribozymes**).

A. Mechanism of Enzyme Action & Activation Energy

Enzymes accelerate biochemical reactions by thousands to millions of times by **lowering the Activation Energy (E_a)** required to achieve the high-energy unstable transition state.

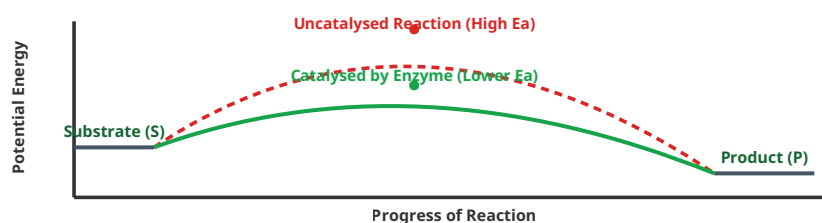
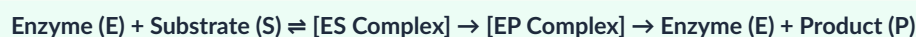


Figure 9.2: Potential energy diagram showing lowering of activation energy by enzyme.



B. Factors Affecting Enzyme Activity & Inhibition

- **Temperature & pH:** Show bell-shaped activity curves with optimum values. Low temperature causes temporary inactivation; high temperature causes irreversible **denaturation**.

- **Substrate Concentration ([S]):** Reaction velocity increases until all active sites are saturated at maximum velocity (V_{max}). K_m (Michaelis constant) = $[S]$ at $V_{max}/2$.
- **Competitive Inhibition:** Inhibitor mimics substrate and binds to active site. **Increases K_m** , while **V_{max} remains unchanged** (e.g., **Malonate** inhibits **Succinic dehydrogenase**).
- **Non-Competitive Inhibition:** Inhibitor binds to allosteric regulatory site. **Decreases V_{max}** , while **K_m remains unchanged**.

C. Classification of Enzymes (6 IUBMB Classes)

Class	Type of Reaction Catalyzed	Representative Reaction Equation	Examples
1. Oxidoreductases	Redox reactions between two substrates	$S(\text{red}) + S'(\text{ox}) \rightarrow S(\text{ox}) + S'(\text{red})$	Cytochrome oxidase, Dehydrogenase
2. Transferases	Transfer of chemical group other than hydrogen	$S - G + S' \rightarrow S + S' - G$	Hexokinase, Transaminase
3. Hydrolases	Hydrolysis of ester, ether, peptide, glycosidic bonds	$A - B + H_2O \rightarrow A - OH + B - H$	Pepsin, Trypsin, Amylase, Lipase
4. Lyases	Cleavage of bonds without hydrolysis leaving double bonds	$X - C - C - Y \rightarrow X - Y + C = C$	Aldolase, Decarboxylase
5. Isomerases	Interconversion of optical, geometric, positional isomers	$\text{Glucose-6-P} \rightleftharpoons \text{Fructose-6-P}$	Triose phosphate isomerase
6. Ligases	Catalyse joining of two molecules with ATP energy	Coupling C-O, C-S, C-N, P-O bonds	DNA Ligase, Glutamine synthetase

D. Co-factors

Holoenzyme (Active Enzyme) = Apoenzyme (Protein Component) + Co-factor (Non-protein Component)

Type of Co-factor	Binding Characteristics	Key NCERT Examples
1. Prosthetic Group	Organic compound tightly and firmly bound to apoenzyme.	Haem in Peroxidase and Catalase (breaks down H_2O_2 to $H_2O + O_2$).
2. Co-enzyme	Organic compound with transient, loose association . Derived from vitamins.	• NAD & NADP contain vitamin Niacin. • FAD contains vitamin Riboflavin (B_2). • Coenzyme A contains Pantothenic acid.
3. Metal Ions	Inorganic coordination ions that bind active site residues and substrate.	• Zn^{2+}: Co-factor for Carboxypeptidase & Carbonic anhydrase. • Mg^{2+}: Co-factor for Hexokinase.